Physical Account (D6) -- 8-Class Resource Ledger, Assumption Register & Counterfactual (1 Page)

1. Subcomponent Mass, Position, Stability & C(5,4) Combinations (Organizer Clarifications Applied)

Stability is computed per subcomponent (x1.0 bay1/bay2, x1.2 hangar/hangar_roof/flight_deck, x3.0 mast_3). Per GRAZIANO_B_COMPOSITION_0855 and GRAZIANO_ROOF_AND_G4_MASS_0904, G4 total mass is 990 kg (140 kg vehicle + 850 kg hangar), G3+G4+G5 = 9,515 kg ~= 9.52 t (5.5 + 0.99 + 3.025 t), and Scenario B installs G1+G3+G4+G5 (9,560 kg, 440 kg mass margin; 9,853.0 kg-eq stability with G5 antenna on hangar_roof, 2,147.0 kg-eq stability margin). Across all C(5,4)=5 sets, 3 exceed 10,000 kg ({G1,G2,G3,G4}=10,035 kg, {G1,G2,G3,G5}=12,070 kg / 12,210 kg-eq, {G2,G3,G4,G5}=13,015 kg / 13,263 kg-eq), while {G1,G2,G4,G5}=7,560 kg (7,853.0 kg-eq roof / 7,898.0 kg-eq mast) and {G1,G3,G4,G5}=9,560 kg (9,853.0 kg-eq roof / 9,898.0 kg-eq mast) pass mass/stability but require ROOF_OUTSIDE_G4_RECOVERY_V1 clearance on hangar_roof because G1 (1,500 W) and G5 (1,200 W) exceed mast_3's single-slot and 2,000 W air limit (2,700 W > 2,000 W).

Guest ID & Role	Subcomponents, Mass & Placement	Stability (kg-eq)	Electrical, Water/Air Thermal & Data Envelope
G1 Passive Radio/Meteo	G1_MAST_SENSOR_HEAD: 45 kg -> mast_3 (0.5 m2, Line A)	135.0 (x3.0)	1.5 kW air (mast_3 <= 2.0 kW; 0 W water); 50 Mbps LAN, 200 kbps SATCOM
G2 Ocean/Weather ROV	G2_ROV_WINCH_CONTAINER_10FT: 3,500 kg -> bay2/bay1_a	3,500.0 (x1.0)	20 kW (300 s winch) / 6 kW cont -> water; 5 Mbps LAN, 500 kbps SATCOM
G3 Seabed AUV+Cradle	G3_AUV_CRADLE_BAY1_20FT: 5,500 kg -> bay1 (Line A)	5,500.0 (x1.0)	1 kW survey / 25 kW (3,600 s) firm charge -> water; 100 Mbps LAN
G4 VTOL Drone+Hangar	G4_DRONE_VEHICLE (140 kg) + G4_HANGAR_STATION (850 kg) = 990 kg -> hangar (Line B)	1,188.0 (x1.2)	35 kW (1,200 s) recovery/charge, 10 kW patrol, 2 kW standby -> water; 20 Mbps LAN
G5 Comms & Edge Compute	G5_COMPUTE_CONTAINER (3,000 kg -> bay2) + G5_MAST_ANTENNA (25 kg -> hangar_roof/mast_3, Line B) = 3,025 kg	3,030.0 roof / 3,075.0 mast	45 kW (43.8 kW water + 1.2 kW air) / 8 kW degraded (7.5 kW water + 0.5 kW air, 30% cap); 1 Gbps LAN, 6 Mbps SATCOM
G6 Civil eDNA Sampler	G6_HYDRO_SKID_10FT: 420 kg -> bay2/bay1_a (ASSUMED, port_only=true)	420.0 (x1.0)	4.5 kW (1,800 s) / 1.5 kW eco -> water; 30 Mbps LAN, 200 kbps SATCOM

2. Mandatory Assumption Register & Code Traceability Table (8 Assumption Families)

Assumption Family	Declared Rule, Units & Provenance Tag	Concrete Files & Code Symbols (file.py::symbol_name)
1. Provenance/mass	Field-level OFFICIAL_BRIEF / ORGANIZER_ERRATA / ASSUMED; G4=990 kg (140+850 kg), G3+G4+G5=9,515 kg ~= 9.52 t, G1+G3+G4+G5=9,560 kg (440 kg margin), 9,853.0 kg-eq (2,147.0 kg-eq margin); G6=420 kg all ASSUMED.	data/overrides/brief_errata.yaml, data/overrides/organizer_clarifications.yaml, src/sdv_governor/manifest.py::parse_guest_manifest, src/sdv_governor/manifest.py::load_guest_catalog
2. Roof & drone clearance	G5_MAST_ANTENNA on hangar_roof (25 kg, 0.3 m2, x1.2) with G4 is governed by ROOF_OUTSIDE_G4_RECOVERY_V1 outside G4_ROTOR_DISC/G4_RECOVERY_CORRIDOR (YES_IF until verified; NO under CONFLICT policy).	data/ship/vessel_baseline.yaml, src/sdv_governor/placement.py::find_feasible_placement, src/sdv_governor/admission.py::evaluate_admission, src/sdv_governor/admission.py::apply_admission_decision_atomically
3. Air & component split	Component-level cooling split: mast_3 <= 2,000 W air (OFFICIAL_BRIEF, G1=1,500 W) and hangar_roof <= 2,000 W air (ASSUMED ASM-ROOF-AIR-2000W, G5 antenna 1,200 W full / 500 W degraded); container 43.8 kW / 7.5 kW water.	data/guests/G5_edge_sigint.yaml, src/sdv_governor/budgets.py::evaluate_eight_class_budgets, src/sdv_governor/models.py::SubcomponentSpec
4. Power & feeders/reservations	200 kW net payload reserve (100 kW post-loss); LINE_A_FWD <= 150 kW (G1+G3), LINE_B_AFT <= 150 kW (G4+G5); separates actual draw vs committed firm reservations (G3 25 kW recharge + G4 35 kW D3 recovery).	data/ship/vessel_baseline.yaml, src/sdv_governor/ship_state.py::FirmReservedLoad, src/sdv_governor/budgets.py::evaluate_eight_class_budgets, src/sdv_governor/degradation.py::execute_graceful_degradation
5. Chilled water & thermal inertia	Linear curve C(T) = 160,000 - 40,000*(T-12)/18 W (160 kW at <=12 deg C, 120 kW at 30 deg C, 72 kW with -40% loop fault); +30 kW x 60 s (1.8 MJ, 600 s recharge) peak buffer never used for sustained (>60 s) loads.	src/sdv_governor/budgets.py::compute_cooling_water_capacity_w, src/sdv_governor/ship_state.py::ThermalInertiaState, src/sdv_governor/degradation.py::evaluate_reconnection
6. State & A.6 transition timing	Causal D3 chain: ACOUSTIC_HOLD cmd -> G3 ACK (15 s nominal, [10, 60] s window) -> 90 s turn/stabilize into wind -> 240 s approach/landing (<= 1,200 s ditching deadline); A.6 60 s cooldown & 15 kW hysteresis.	src/sdv_governor/arbitration.py::arbitrate_live_conflict, src/sdv_governor/replay.py::_process_acoustic_hold_ack_event, src/sdv_governor/replay.py::_apply_due_d3_milestones, src/sdv_governor/scheduling.py::evaluate_a_prime_window
7. Compute rack & data links	4 nodes (node_1..node_4), 32c / 128 GiB / 1 acc each (80% allocatable = 25c / 102,400 MiB per node, max 3 discrete accelerators, 16 TB storage); 10 Gbps LAN (<=1 Gbps/port), SATCOM 10/6/2/0 Mbps (W4 EMCON buffers G5).	data/ship/vessel_baseline.yaml, src/sdv_governor/budgets.py::evaluate_eight_class_budgets, src/sdv_governor/placement.py::find_feasible_placement
8. Priorities & degraded capability	Hard physical/safety constraints first; lexicographic mission score (survey_primary lands T=1,095 s, rescue_primary lands T=855 s); G5 step-down (45 -> 8 kW) retains 30% capability (capability_retention_ratio=0.30).	data/priorities/survey_primary.yaml, data/priorities/rescue_primary.yaml, src/sdv_governor/arbitration.py::_evaluate_candidate_feasibility_and_score, src/sdv_governor/degradation.py::execute_graceful_degradation

3. Scenario B (G1+G3+G4+G5) Power, Feeder, Water & Air Ledger Across Transitions

* Electrical Power (actual / committed): Initial 57.5 / 81.5 kW (G1=1.5, G3=1.0 actual / 25.0 firm charge, G4=10.0, G5=45.0) -> 82.5 / 106.5 kW at T=480 s when G4 35.0 kW recovery/charge is reserved -> at T=720 s (100 kW generator reserve), 106.5 kW > 100 kW triggers G5 step-down (45.0 -> 8.0 kW), yielding 45.5 / 69.5 kW <= 100 kW (30.5 kW committed margin).
* Chilled-Water & Air Domains: Water load is 54.8 / 78.8 kW initial -> 79.8 / 103.8 kW with G4 35 kW -> 43.5 / 67.5 kW <= 72 kW after G5 step-down (43.8 -> 7.5 kW water), surviving 30 deg C + -40% loop loss (72 kW). Air cooling is 1,500 W on mast_3 (G1) + 1,200 W on hangar_roof (G5 antenna) = 2,700 W initial (2,000 W after G5 step-down to 500 W roof air).
* Feeder Routing (<= 150 kW/line): LINE_A_FWD = 2.5 / 26.5 kW (G1 1.5 kW + G3 1.0 / 25.0 kW); LINE_B_AFT = 80.0 -> 43.0 kW (G4 35.0 kW + G5 45.0 -> 8.0 kW).

4. What would change if the vessel had one single platform modification (mast_4)?

If the vessel added one second mast slot mast_4 (<= 60 kg, <= 0.6 m2, <= 2,000 W independent forced-air cooling, x3.0 stability factor), before vs. after solver results change as follows: for {G1, G3, G4, G5} (9,560 kg) and {G1, G2, G4, G5} (7,560 kg), G5_MAST_ANTENNA (25 kg, 0.3 m2, 1,200 W air) no longer collides with G1 (45 kg, 0.5 m2, 1,500 W air) on mast_3 (2,700 W > 2,000 W) and no longer requires hangar_roof placement or conditional G4 rotor-clearance verification (ROOF_OUTSIDE_G4_RECOVERY_V1), moving from 9,853.0 kg-eq (hangar_roof) to 9,898.0 kg-eq (mast_4, 2,102.0 kg-eq margin) with unconditional static placement (YES at port; YES_IF under 14-day Mission Envelope A.10 solely for W4 EMCON offline buffering and W2/W3/W5 time-multiplexed windows). However, mass (10,000 kg), bay capacities (bay1 <= 8,000 kg, bay2 <= 4,000 kg), and stability (12,000 kg-eq) still limit the other three 4-guest combinations: {G1,G2,G3,G4} (10,035 kg), {G1,G2,G3,G5} (12,070 kg, 12,210 kg-eq), and {G2,G3,G4,G5} (13,015 kg, 13,263 kg-eq, plus G2+G5 = 6,500 kg > 4,000 kg in bay2).